

Literature Review: Optimal Sensor Placement for Wastewater Network Monitoring

Source classification supporting Project Objective PO1 and Deliverable D1

Project: Optimal Sensor Placement for Wastewater Network Monitoring, SA Water
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Abstract

This review supports Project Objective PO1 and Deliverable D1. It examines six sources spanning four bodies of work relevant to the project: the formulation of the sensor placement problem itself, the identification of the network attributes that determine where monitoring is most warranted, the practical reliability of the sensor hardware once installed, and the broader evidence base on AI-driven sewer condition assessment against which the project's parameter catalogue can be cross-checked. Each source is classified by optimisation formulation, objective function, input parameters, network type, and applicability to a gravity trunk sewer network. Three placement formulations are compared: deterministic global optimisation with decoupled hydraulics, topological ranking by in-relevance harmonic centrality, and data-driven sparse sensing using singular value decomposition and QR factorisation with column pivoting. Two treatments of sewer condition are examined, a state-of-the-art factor review and a data-driven Bayesian network, and one field study of paired flow and level sensors tests the reliability assumption that the placement formulations leave implicit. The reviewed placement formulations are reusable, but their objective functions are not: none targets overcapacity arising from growth-related load increase, which is the project's primary application. Six gaps are recorded, of which the absence of an agreed observability measure for rising sewer level under increased connected load is the principal open question with the client.

Keywords: wastewater network; sewer monitoring; optimal sensor placement; sparse sensing; complex network theory; Bayesian networks; sensor reliability; SA Water.

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1 Scope and Purpose

This review supports Project Objective PO1 and Deliverable D1. It examines six sources that together span four bodies of work relevant to the project: the formulation of the sensor placement problem itself, the identification of the network attributes that determine where monitoring is most warranted, the practical reliability of the sensor hardware once installed, and the broader evidence base on AI-driven sewer condition assessment against which the project’s parameter catalogue can be cross-checked. Each source is classified by optimisation formulation, objective function, input parameters, network type, and applicability to a gravity trunk sewer network, consistent with the classification scheme committed to in PO1.

The six sources are Simone et al. (2023), which compares a deterministic optimisation formulation against a topological one for monitoring planning in sewer networks; Ding et al. (2026), which formulates sensor placement as a data-driven sparse sensing problem for system-wide flow reconstruction in an urban drainage network; Malek Mohammadi et al. (2020), a state-of-the-art review of the factors influencing sewer pipe condition; Ma et al. (2025), a data-driven Bayesian network model quantifying the influence of those factors on defect-based pipe condition; Alshami et al. (2023), a field study of flow and level sensor performance in small- and medium-diameter sewers in Hong Kong; and Ma et al. (2026), a systematic review of 71 studies (2016–2025) on AI-driven sewer condition assessment from both structured asset data and CCTV imagery. The first two inform the placement method (D3); the third and fourth inform the placement parameter catalogue (D2); the fifth bears on sensor type selection and the cost and information trade-off analysis (D5), and on the reliability assumptions embedded in the placement objective itself; the sixth cross-checks the parameter catalogue (D2) against a much larger evidence base and surfaces explainability techniques relevant to the project’s explainability requirement (R4).

2 Formulations of the Sensor Placement Problem

2.1 Deterministic optimisation

Simone et al. (2023) formulate sensor placement as a mixed-integer problem solved to global optimality using the SCIP global optimisation solver. Each node of the network carries a binary installation variable, a second binary variable records whether a contaminant is detected at a node under a given injection scenario, and a cardinality constraint fixes the total number of sensors. The objective maximises network reliability, defined as the ratio of detected node-scenario pairs to the total number of detectable pairs across all scenarios.

The methodological feature most relevant to this project is the decoupling of the optimisation from the hydraulic model. Flow velocities and pollutant concentrations are computed once in SWMM and enter the optimisation as fixed parameters, so the solver does not invoke a hydraulic simulation at each candidate configuration. This is the same architectural choice that makes the project’s own proposed approach tractable within the project timeframe: simulation is used to characterise the network, not to search it. Reported solve times were under one second even for the larger of the two test networks.

2.2 Topological optimisation via complex network theory

The second formulation in Simone et al. (2023) abstracts the network as a directed graph and ranks nodes by in-relevance-based harmonic centrality, a variant of harmonic centrality that weights each node by its intrinsic relevance (here, lateral inflow) and that accumulates only information arriving at a node. The justification offered for harmonic centrality in this setting is that transport in a gravity sewer follows shortest paths determined by slope, so the metric’s underlying assumption matches the physics of the system. No simulation is required; only topology and inflow data are needed.

Two results bear directly on the project. First, the ranking is normalised to a $[0, 100]$ range and the authors adopt values above 20 as the criterion for candidate sensor locations, which yields a small candidate set on a real network. Second, head nodes return a null metric value, because information does not arrive at them, and are therefore excluded as candidates. This second property is a material limitation for the project’s target application and is discussed in Section 6.

2.3 Data-driven sparse sensing via reduced-order reconstruction

Ding et al. (2026) formulate sensor placement as a data-driven sparse sensing (DSS) problem, coupling EPA-SWMM simulation with singular value decomposition (SVD) and QR factorisation with column pivoting to select the smallest set of nodes from which system-wide flow can be reconstructed. An SVD of a simulated training ensemble (250 scenarios spanning ten rainfall events and five imperviousness and five infiltration settings) yields a reduced temporal basis; QR factorisation with column pivoting on that basis then identifies the nodes whose measurements most efficiently reconstruct the full network state. As in Simone et al. (2023), the hydraulic model is used once, offline, to characterise the network rather than being re-invoked at each candidate configuration, the same architectural choice that keeps the approach tractable and that the project’s own proposed method also relies on.

Applied to a 77-node storm sewer network in Duluth, Minnesota, three DSS-selected sensors reconstructed system-level flow with a mean Nash-Sutcliffe efficiency (NSE) of 0.949 across five observed storm events, rising to 0.993 at ten sensors; reconstruction of a 200-year design event improved from an NSE of 0.842 at one sensor to 0.994 at ten. Benchmarked against an exhaustive search (feasible for one to four sensors) and against 20,000-sample Monte Carlo random placements (five to ten sensors), the DSS-selected configurations tracked the global optimum closely (mean NSE gap of 0.002–0.013) and substantially outperformed random placement, whose reconstructions were frequently and severely negative in NSE. The framework degraded gracefully under multiplicative Gaussian measurement noise of up to 15% and, at moderate to high sensor counts, under single-sensor dropout, though dropout sensitivity was concentrated in a small number of structurally critical sensors whose importance was better explained by their redundancy, post-dropout conditioning and modal loading than by their QR selection rank alone.

Two aspects of this formulation are directly relevant to the project’s method (D3) and to Deliverable D5. First, DSS supplies an evaluation measure, reconstruction NSE against a held-out event, that is a genuine measure of network observability rather than a topological proxy, which is the gap Section 6 identifies as unresolved in the other placement formulations reviewed here. Second, the robustness testing built into the DSS evaluation, specifically its treatment of measurement noise and of sensor dropout as first-class experimental conditions, is the only placement formulation reviewed so far that evaluates a placement’s fault tolerance rather than assuming reliable readings at every instrumented node once installed; this partially closes, without fully resolving, the reliability gap raised in Section 5 below. The authors are explicit, however, that DSS is a linear reduced-order framework and that this limits its ability to represent the threshold-driven behaviour of overflow-prone nodes, which sit at or near zero flow under ordinary conditions and activate abruptly once a hydraulic threshold is exceeded; this is precisely the class of node the project’s target application is designed to monitor, and the authors identify it as the framework’s most significant limitation for exactly this reason.

2.4 Comparative findings

The two formulations were applied to a benchmark network of 77 nodes, 79 links and one outfall, and to a real network of 531 nodes, 530 pipes and one outfall. On both, the node sets selected by the solver fell within the range of nodes ranked highest by the topological metric, or were adjacent to them, and lay along the same principal flow paths. The outfall was the highest-ranked node

in both cases, as expected.

The two approaches diverge in one respect that matters for sensor selection. The optimisation solution is sensitive to the assumed sensor detection threshold: relaxing the threshold from 0.003 to 0.01 mg/L increased the number of sensors required for full coverage from four to eleven on the real network. The topological ranking, depending only on structure and inflow, does not change with sensor quality at all. For a project that must compare float switches against level sensors under an agreed observability measure (PO5), this is a direct finding: a purely topological criterion cannot by itself discriminate between sensor modalities, and some threshold- or capability-dependent component is required alongside it.

The authors conclude by proposing a two-stage procedure in which the topological metric pre-screens candidate nodes and the deterministic optimisation searches within that reduced set, on the grounds that deterministic solvers become computationally demanding at scale. They also note that the topological approach demands competent interpretation, since its results must still be checked for hydraulic plausibility.

A third point of comparison follows from placing Ding et al. (2026) alongside the two formulations above. Where the deterministic and topological methods in Simone et al. (2023) both target contaminant or intrusion detection and evaluate placement quality by a reliability or centrality score, DSS targets flow reconstruction and evaluates placement quality directly against held-out flow observations. This makes DSS’s NSE-based evaluation the closest of the three to a genuine observability measure, but it inherits a different limitation: as a linear method, it is best suited to nodes whose flow varies continuously, and it is weakest exactly at the threshold-crossing overflow nodes the project’s overcapacity application must monitor.

3 Factors Influencing Sewer Condition

Malek Mohammadi et al. (2020) review condition prediction models published between 2001 and 2019 and catalogue the variables used in them. Following Davies et al., the factors are grouped as physical (age, material, diameter, length, depth, slope, shape, joint type, invert elevations), environmental (soil type, corrosivity, moisture, groundwater level, bedding and backfill, road type, traffic, root interference) and operational (blockages, flow velocity, sediment level, infiltration and exfiltration, sewer type, maintenance history). Of the twenty independent variables surveyed, pipe age, diameter and length emerge as the most frequently significant contributors to deterioration.

Three observations from this review carry into the parameter catalogue (D2).

3.1 Several factors are contested rather than settled

Diameter, depth, slope and pipe length all appear with contradictory signs across studies. Larger diameters are associated with both lower risk (attributed to more careful design and installation of larger assets) and higher risk (attributed to greater surface area in contact with sewage and soil). Depth is insignificant in the majority of the models reviewed. The review’s own explanation is that condition models are fitted to local geography, so a factor’s direction of effect is not transferable between utilities. For this project, the implication is that the catalogue should record each parameter’s mechanism and its evidential status, not a single assumed sign.

3.2 Data availability biases which factors are studied

Physical attributes dominate the literature because they are cheap to collect and already held in asset inventories; environmental and operational data are frequently absent. The review recommends greater use of environmental and operational variables, specifically soil type, bedding material, flow rate and soil corrosivity. This mirrors the data-availability constraint the project itself faces and supports documenting availability status explicitly for each catalogued parameter.

3.3 Condition is assessed against defined rating protocols

The review distinguishes structural condition (defects, strength, resistance to external load) from operational condition (ability to meet service requirements), and notes that structural results drive rehabilitation decisions whereas operational results drive cleaning and maintenance. Rating protocols include WRc, PACP, the NRC protocol and, relevant to an Australian deployment, the WSAA standard, with condition typically graded from 1 to 5.

3.4 A larger and more recent evidence base

Ma et al. (2026) extend this picture with a systematic review of 71 peer-reviewed studies published between 2016 and 2025 on AI-driven sewer condition assessment, roughly triple the span covered by Malek Mohammadi et al. (2020). Restricting attention to the subset of that review using structured, non-visual data, that is, the traditional machine-learning studies most comparable to condition-prediction practice, pipe age and diameter recur as the two most frequently identified influential factors, cited in 54.2% and 25.0% of the reviewed studies respectively, followed by pipe material, length and slope; this corroborates, at a larger sample size, both the age and diameter findings of Malek Mohammadi et al. (2020) and the mutual-information ranking reported by Ma et al. (2025). The review also confirms the data-availability bias already noted above: physical attributes appear in every study surveyed, geospatial and socio-environmental factors in roughly half, and hydraulic or chemical properties, the category closest to the flow and level behaviour this project needs to monitor, in under one fifth. For the parameter catalogue (D2), this strengthens the case for treating pipe age, diameter and material as a well-evidenced core, while confirming that growth-related load and flow behaviour remain comparatively under-studied and therefore require the project’s own analysis rather than a literature-derived weighting.

Two further points from Ma et al. (2026) carry into the placement method (D3) and the explainability requirement (R4). First, the review’s summary of standardised inspection protocols, PACP, MSCC, WSA 05, HKCCEC, CSA PLUS 4012–10 and EN 13508–2 among others, converges on a five-point condition scale in which grade 4 marks the threshold for required intervention and grade 5 marks imminent or actual failure; this scale is a candidate reference point for expressing the project’s own placement priority in familiar, protocol-aligned terms when presenting recommendations to the asset management team, rather than in raw topological or observability units that carry no meaning outside the project. Second, the review’s discussion of interpretability, in particular SHapley Additive exPlanations (SHAP) for quantifying factor contributions and Bayesian or Monte Carlo dropout methods for attaching confidence intervals to a prediction, is offered there as a remedy for the low transparency of machine-learning condition models; the same techniques are a plausible route for the project to attach a defensible confidence statement to a placement recommendation once the parameter catalogue (D2) is finalised, directly supporting the explainability requirement (R4) that any recommendation must be understandable to a non-specialist approver.

4 Quantifying Factor Influence: Bayesian Network Modelling

Ma et al. (2025) address the disagreement identified above by estimating factor influence from data rather than from domain assumption. Fourteen influence factors, spanning pipe physical properties, environmental context and climate, were assembled into an integrated database for the Hong Kong sewer network, which spans approximately 4,700 km. CCTV defect records were joined to GIS pipe records on pipe identifier, and environmental and climate attributes were attached by spatial join; approximately 23,000 pipe records remained after cleansing.

The target variable is an aggregated defect-based condition on a six-level scale, derived by taking the maximum of the internal condition grade, the service condition grade, and a structural performance grade that adjusts the internal grade for installation age and slope location. Network

structure was learned by a score-based Bayesian search using the Bayesian Information Criterion, producing a directed acyclic graph of 15 nodes and 31 links in which pipe condition is the child of pipe age, pipe diameter and population.

Factor influence was then ranked by three sensitivity measures: mutual information, joint probability, and true risk influence, the last computed by forcing each state of a factor to certainty and taking the mean of the resulting highest and lowest deviations from the prior. All three agree on the leading factors. By mutual information, pipe age ranks first (0.124), followed by diameter (0.032), population (0.027) and soil type (0.024); every remaining factor scores below 0.02. Scenario analysis identified the combination most associated with critical condition as age above 50 years, diameter below 200 mm, population density below 15,000, and fill or granitic soils; forcing this combination raised the probability of critical condition from 17% to 72%. Validation against a withheld inspection record returned the correct condition class with 85% probability.

Two aspects of the method are transferable regardless of whether the numerical findings are. First, the Bayesian network handles missing data and supports both forward and backward inference, which suits a setting where some network attributes are known to be incomplete. Second, the sensitivity measures provide a defensible basis for reducing a long candidate parameter list to a working subset, which is precisely the task PO2 sets.

5 Sensor Technology Performance in Small-Diameter Networks

The three sources above all treat a sensor reading as ground truth once a node is instrumented: the placement problem is where to put sensors, not whether a sensor at a given node can be trusted. Alshami et al. (2023) test that assumption directly. Ultrasonic flow sensors and ultrasonic level sensors were installed together in 12 maintenance holes across four districts of the Hong Kong sewer network, spanning inlet pipe diameters of 0.15–0.6 m, and monitored continuously for one year (June 2019 to June 2020), yielding approximately half a million observations.

5.1 Reliability findings

The flow sensors performed poorly against two independent checks. The correlation coefficient between flow-sensor velocity and level-sensor depth across all maintenance holes was 0.36, and it varied widely by location, from 0.92 at one maintenance hole to -0.13 at another. Flow rates computed from the flow sensors were compared against flow rates computed from level-sensor depth via Manning’s equation, and the average relative error across all maintenance holes was 72.14%, with individual maintenance holes ranging from 19% to 97.6%. At one maintenance hole, the flow sensor recorded velocities exceeding 30 m/s while the level remained essentially constant, an implausible reading for a gravity sewer that the authors attribute to sensor malfunction under a specific combination of conditions rather than to genuine flow.

5.2 Causes identified

Three design and operational conditions were found to degrade flow-sensor accuracy, each traced by comparing maintenance holes that performed well against maintenance holes that did not.

Outlet pipe diameter. The ultrasonic flow sensors measure velocity by firing a pulse across the flow surface and timing its reflection; this depends on a consistent flow direction. In small-diameter outlet pipes the flow direction becomes inconsistent as sewage exits the maintenance hole, and the smaller the pipe, the more pronounced the effect. Two maintenance holes with otherwise identical design (single inlet and outlet) but outlet diameters of 0.15 m and 0.225 m showed markedly different reliability, with the narrower pipe performing worse.

Sewage level relative to the outlet pipe. Once the sewage level rises above the outlet pipe’s cover level, the dominant motion at the sensor becomes vertical rather than horizontal, which the horizontal-flow sensor cannot resolve. Readings in this regime were either near zero or erratic, including the implausible high-velocity readings noted above. Because sewage level can exceed the outlet cover level from ordinary high usage as well as from a blockage, this failure mode also produces false positives for blockage and overflow detection specifically at the moments when detection matters most.

Multiple inlet and outlet pipes with varying invert levels. Ten of the twelve maintenance holes studied had two or three inlet pipes. Where a secondary inlet contributed materially to inflow, and particularly where inlet pipes had different downstream invert levels, flow direction at the sensor became inconsistent and accuracy fell. Where a secondary inlet’s contribution was small relative to the main pipe, accuracy was largely unaffected, indicating that inlet count alone is not the determining factor; relative inflow contribution and invert-level variation are.

5.3 Proposed alternative and its relevance

Because level sensors alone remained reliable throughout, the authors propose deriving a vertical velocity from successive level readings (five minutes apart in the study) rather than relying on the horizontal flow sensor. A rising level with consistently positive vertical velocity signals a blockage; the magnitude and sign of the vertical velocity, combined with maintenance hole geometry (cover level, invert level, pipe diameter), allow a simple estimate of remaining time to overflow. Validated against the same twelve maintenance holes, this measure tracked the observed sewage level far more closely than the flow sensors did, and requires only the cheaper of the two sensor types.

Three implications follow for this project. First, the reliability of a sensor at a candidate node is not uniform across the network: it depends on outlet diameter, on how often level is expected to exceed the outlet cover (itself a function of the growth-driven load the project is designed to address), and on the number and configuration of connected pipes at that maintenance hole. Second, the finding motivates why level sensors, not flow sensors, are the more dependable instrument in exactly the small-diameter, multi-inlet conditions common at the periphery of a growing network, which is also where the project’s target overflow mechanism concentrates. Third, the vertical-velocity construction is a candidate observable for the network visibility measure the project has not yet defined: unlike centrality, which is purely topological, it is a directly measurable quantity tied to the physical event (rising level) the project needs to detect.

5.4 Corroboration from adjacent inspection technologies

Ma et al. (2026) broaden this picture beyond flow and level sensors. Reviewing laser profiling and LiDAR as alternatives to CCTV for internal sewer inspection, they note that both methods offer higher data resolution and acquisition efficiency than CCTV but remain susceptible to environmental interference, point-cloud noise from surface reflection or debris, and, notably, a reliable operating range limited to above the water level, the same operating boundary that Alshami et al. (2023) identify as the point at which flow-sensor accuracy for the twelve Hong Kong maintenance holes collapses. That the same limitation recurs across an entirely different sensing modality strengthens rather than merely repeats the reliability finding above: any technology that measures at or below the design water surface, whether by ultrasonic flow, laser range-finding or LiDAR, degrades once the surface itself is disturbed by a rising or turbulent flow, which is precisely the operating condition the project’s overcapacity application must observe most closely.

6 Synthesis and Gaps

Tables 1 and 2 classify all six sources against the PO1 scheme, split across two tables for type-setting. Six gaps follow from this set of sources.

Table 1: Classification of reviewed sources against the PO1 scheme (i).

	Simone et al. (2023)	Malek Mohammadi et al. (2020)	Ma et al. (2025)	Alshami et al. (2023)
Formulation	(a) MINLP, global solver; (b) centrality ranking	Narrative review; no optimisation	Bayesian network, score-based structure learning	Empirical field study; no optimisation or prediction model
Objective function	(a) Maximise detection reliability over injection scenarios; (b) none — ranking only	Not applicable	Posterior probability of defect-based condition class	Not applicable; reports correlation and relative error against Manning’s equation
Input parameters	(a) Topology, flow velocity, concentration (SWMM-derived), detection threshold; (b) topology, nodal inflow, slope	20 physical, environmental and operational attributes	14 factors: physical, environmental, climate; CCTV defect records	Maintenance hole design criteria (pipe diameters, invert levels, inlet/outlet count); paired flow and level readings
Network type	Gravity sewer, directed graph; 77-node benchmark and 531-node real network	Gravity sanitary, storm and combined sewers	Municipal sewer network, approx. 4,700 km	Small- and medium-diameter gravity sewer, 12 maintenance holes, Hong Kong
Purpose	Contaminant / illicit intrusion detection	Structural and operational deterioration	Defect-based condition and inspection prioritisation	Sensor reliability for blockage and overflow detection
Applicability to trunk sewer	High: directed-graph abstraction and decoupled hydraulics both transfer	Moderate: parameter taxonomy transfers; effect directions do not	Moderate: method transfers; coefficients are specific to Hong Kong	High: diameter range and maintenance hole configurations overlap with SA Water’s network

6.1 Objective mismatch

The condition- and placement-oriented studies reviewed above, together with Ding et al. (2026), address contaminant detection, deterioration, or general flow reconstruction. None targets over-capacity or overflow arising from growth-related load increase, which is the project’s primary application. Ding et al. (2026) comes closest, since flow reconstruction is a genuine hydraulic quantity rather than a topological proxy, but as a linear reduced-order method it is explicitly weakest at the threshold-crossing behaviour that defines an overflow event, and its reconstruction objective, minimising deviation from a simulated flow series, is not the same objective as detecting when a level exceeds a critical threshold. The placement formulations reviewed are reusable, but their objective functions are not: an observability measure for rising sewer level under increased connected load must be defined rather than adopted.

6.2 Head nodes and upstream risk

The topological metric in Simone et al. (2023) assigns null relevance to head nodes and therefore excludes them. For contaminant detection this is correct, since a head node receives nothing. For

Table 2: Classification of reviewed sources against the PO1 scheme (ii).

	Ding et al. (2026)	Ma et al. (2026)
Formulation	Data-driven sparse sensing: SVD for a reduced temporal basis, QR factorisation with column pivoting for sensor selection	Systematic literature review (71 studies); taxonomy of traditional ML and deep-learning models, not itself an optimisation formulation
Objective function	Maximise system-level flow-reconstruction accuracy (Nash-Sutcliffe efficiency) against held-out SWMM-simulated events	Not applicable; reports the classification, detection or segmentation performance of the models surveyed
Input parameters	SWMM-simulated flow time series across an ensemble of rainfall and hydrologic parameter scenarios; no explicit topology or attribute inputs beyond the SWMM model itself	ML: physical, geospatial, socio-environmental, operational, hydraulic/chemical and climatic factors. DL: CCTV imagery, and laser, LiDAR or point-cloud data
Network type	Storm sewer, 77-node network, Duluth, Minnesota	Aggregated across 71 studies and multiple countries and utilities; no single network
Purpose	System-wide flow reconstruction from sparse measurements, for flood monitoring	Structural/operational condition prediction (ML) and CCTV-based defect classification, detection and segmentation (DL)
Applicability to trunk sewer	Moderate: the offline-basis architecture transfers, and the reconstruction-NSE evaluation measure is directly reusable, but the linear framework’s weakness at threshold-driven nodes is a poor match for overflow-focused monitoring	Moderate: the factor taxonomy and pipe-age/diameter evidence transfer directly to the parameter catalogue (D2); the review addresses condition assessment rather than placement, so its model formulations do not transfer to D3

growth-driven overcapacity it may be the wrong exclusion, because new residential connections attach at the periphery of the network and the resulting surcharge can originate upstream. Any centrality-based objective adopted for this project must be tested against this case before use.

6.3 Load representation

Both Malek Mohammadi et al. (2020) and Ma et al. (2025) treat population or population density as a static environmental attribute of a pipe. The project requires the opposite framing: load as a quantity that changes when a development is connected. No source reviewed so far models placement under a changing load distribution, and this appears to be the sharpest gap in the reviewed set.

6.4 Absence of a common evaluation measure

Simone et al. (2023) evaluates against detection reliability, Ma et al. (2025) against posterior condition probability, and the models surveyed in Malek Mohammadi et al. (2020) against condition-rating accuracy. None is directly a measure of network observability. Ding et al. (2026) evaluates against reconstruction NSE, the closest of the sources reviewed to a direct observability measure, and the standardised condition grades summarised in Ma et al. (2026) offer a structural/operational reference scale, but neither is a measure of overflow observability under growth-driven load: NSE measures how well an unobserved flow series can be reconstructed from sparse measurements in general, not whether a rising level relative to a specific overflow threshold would be detected in time to act, and the condition grade measures asset deterioration, not real-time monitoring coverage. This is consistent with the absence of a unified quantitative evaluation process noted elsewhere in the sensor placement literature, and it reinforces the operational definition of network visibility as the project’s principal open question with the client.

6.5 Uniform sensor reliability is assumed, not tested

Both formulations in Simone et al. (2023), and by extension any placement method built the same way, treat detection at an instrumented node as reliable once installed; the optimisation and the centrality ranking differ only on where to place sensors, not on whether a given node can be trusted to report correctly. Alshami et al. (2023) show this assumption fails in small-diameter, multi-inlet maintenance holes, exactly the configuration expected at the periphery of the network where growth-driven connections attach. A placement method for this project should therefore weight candidate nodes by expected in-situ reliability, informed by outlet diameter, inlet configuration, and expected level relative to the outlet cover, rather than by topological or hydraulic importance alone. This also bears on sensor type selection (R8): the evidence favours a level-sensor-led design over a flow-sensor-led one specifically in the small-diameter, multi-inlet locations most likely to host new connections. Ding et al. (2026) is a partial exception in that it tests robustness to measurement noise and to sensor dropout directly, but both are properties of the reconstruction algorithm’s tolerance to missing or noisy data, not of the physical sensor malfunction modes Alshami et al. (2023) document; a placement method for this project should therefore weight candidates by both kinds of reliability, algorithmic robustness to missing readings and physical robustness of the reading itself, rather than treating either as a stand-in for the other.

6.6 No standard link to the condition-grade scale asset managers already use

Ma et al. (2026) show that a five-point structural/operational condition scale, with intervention triggered at grade 4 and failure at grade 5, is the practitioner-facing convention across the protocols it surveys (PACP, MSCC, HKCCEC and others). None of the placement formulations reviewed here, Simone et al. (2023) and Ding et al. (2026), express their output in that scale, nor do the condition-modelling sources Malek Mohammadi et al. (2020) and Ma et al. (2025). Presenting a placement recommendation (D3) or business case (D7) in a topological score, an NSE value, or a posterior probability risks the same approval-stage failure a non-specialist funding approver would reject outright; a translation from whatever observability measure the project adopts into a grade-4-equivalent priority threshold is worth defining early, not left to the write-up.

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